

Introduction to the Course 'Advanced PyTorch Techniques and Applications'

Welcome to the course **Advanced PyTorch Techniques and Applications**. The third part of the 3 course Specialization in [PyTorch Ultimate 2024 - From Basics to Cutting-Edge](#)

The course is structured into twelve comprehensive modules:

- Recommender Systems:** In this module, we will explore the basics of recommender systems, starting from foundational concepts and progressing through hands-on coding exercises. You'll create datasets, develop and train models, and learn how to incorporate user and item information for improved recommendations. Finally, we will implement evaluation metrics to measure the system's performance.
- Autoencoders:** In this module, we will dive into autoencoders, covering both theoretical aspects and practical implementations. You will gain a solid understanding of how autoencoders work, their applications, and get hands-on experience coding these models.
- Generative Adversarial Networks:** In this module, we will cover the essentials of generative adversarial networks, including an overview of their principles and coding implementations. You will learn to develop a GAN model and engage in exercises that challenge you to apply these techniques to specific tasks.
- Graph Neural Networks:** In this module, we will explore graph neural networks, starting with the basics and moving through coding implementations. You'll learn how to prepare data, train models, and evaluate their performance, all within the context of GNNs.
- Transformers:** In this module, we will delve into Transformers, beginning with foundational concepts and then focusing on their application to vision tasks. You'll gain hands-on experience in implementing and training a Vision Transformer on a custom dataset.
- PyTorch Lightning:** In this module, we will introduce you to PyTorch Lightning, a powerful framework for PyTorch model development. You'll learn the basics, implement models, and explore techniques such as early stopping to optimize your training processes.
- Semi-Supervised Learning:** In this module, we will cover semi-supervised learning, beginning with foundational concepts and progressing through practical implementations. You will learn about supervised reference models, set up datasets, and develop models that effectively utilize both labeled and unlabeled data.
- Natural Language Processing (NLP):** In this module, we will explore the vast field of Natural Language Processing, from fundamental concepts to hands-on coding implementations. You'll learn to work with word embeddings, sentiment analysis, pre-trained models, and advanced topics like zero-shot classification and vector databases.
- Miscellaneous Topics:** In this module, we will cover a range of miscellaneous topics in machine learning, including architectures like ResNet and Inception, and concepts such as Extreme Learning Machines. Each topic will include both theoretical understanding and practical coding exercises.
- Model Debugging:** In this module, we will focus on model debugging techniques, specifically using hooks. You'll learn the theoretical aspects and get hands-on experience implementing hooks to troubleshoot and optimize your models.
- Model Deployment:** In this module, we will explore the essentials of model deployment, covering both on-premise and cloud-based strategies. You'll learn to deploy models using Flask, consume data from APIs, and utilize Google Cloud for deploying model weights and REST APIs.
- Final Section:** In this module, we will conclude the course by summarizing key concepts and techniques covered throughout. Additionally, we will provide resources and recommendations for further learning to help you continue your journey in advanced PyTorch techniques and applications.

Target Learner: This course is designed for data scientists, machine learning engineers, and AI researchers with a solid foundation in PyTorch. Prerequisites include a strong understanding of machine learning fundamentals, proficiency in Python programming, and prior experience with PyTorch.

Learning Objectives:

- Implement and train recommender systems using PyTorch.
- Develop and train autoencoders for data compression and feature extraction.
- Explore and implement Generative Adversarial Networks (GANs) for creative applications.
- Understand and work with Graph Neural Networks (GNNs) for graph data.
- Utilize advanced architectures like Transformers and Vision Transformers for various tasks.
- Learn about model deployment, debugging techniques, and semi-supervised learning.

Go to next item

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Completed

